Tarun Kakarala

Cartography & GIS Portfolio

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Maps produced in QGIS and Google Earth Engine

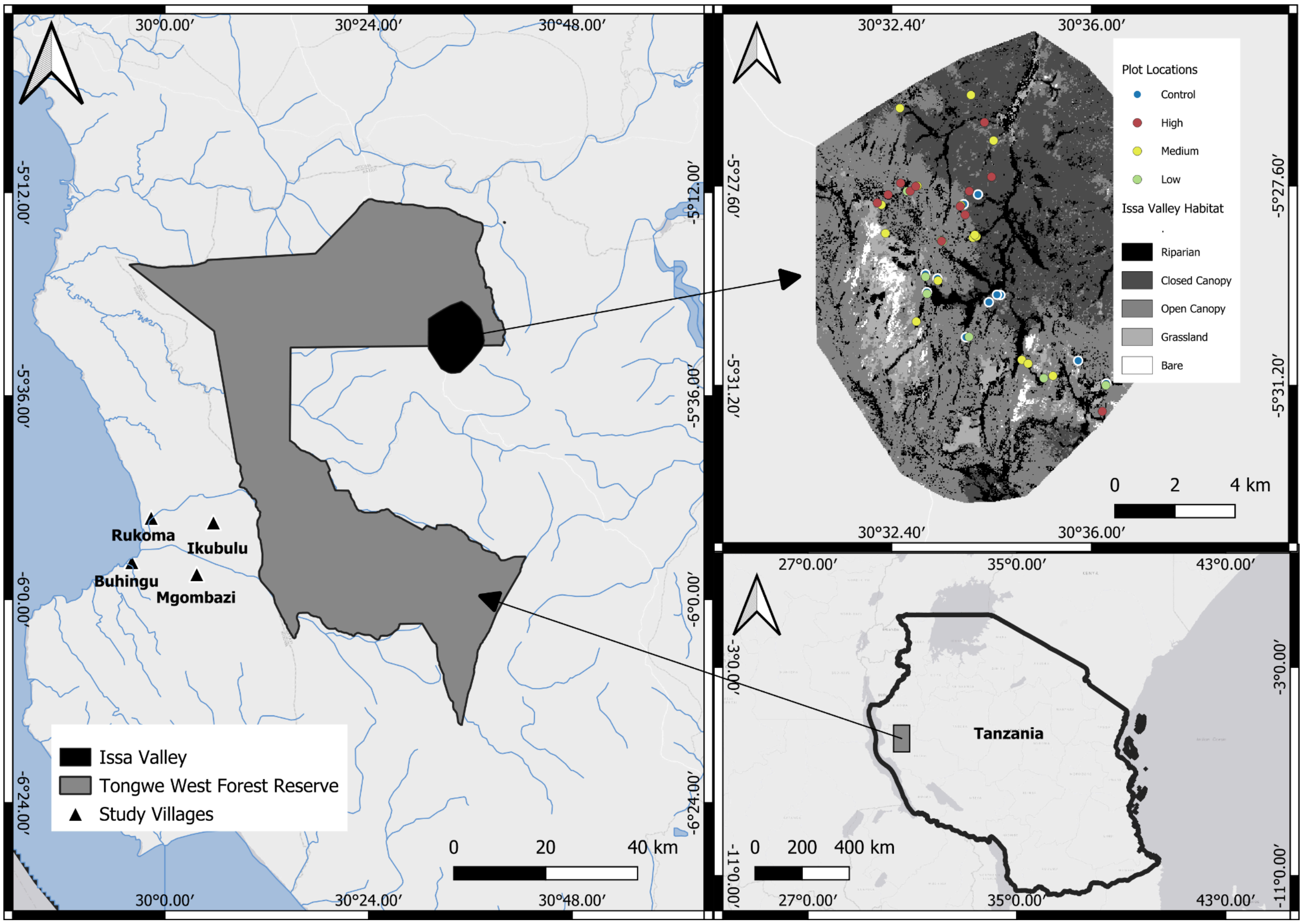
This portfolio presents selected cartographic work from field research in the Greater Mahale Ecosystem, western Tanzania. The three maps span scales from regional context to a fine-scale study of herder movement in relation to habitat and terrain to a two-decade land-cover change time series. Together they demonstrate the full workflow behind a thematic map series: satellite-derived classification in Google Earth Engine, integration of field-collected spatial data, and cartographic production in QGIS.

01 Agropastoral Landscape in the Greater Mahale Ecosystem

02 Agropastoral Movement and Use of the Landscape

03 Land Cover Change Time Series

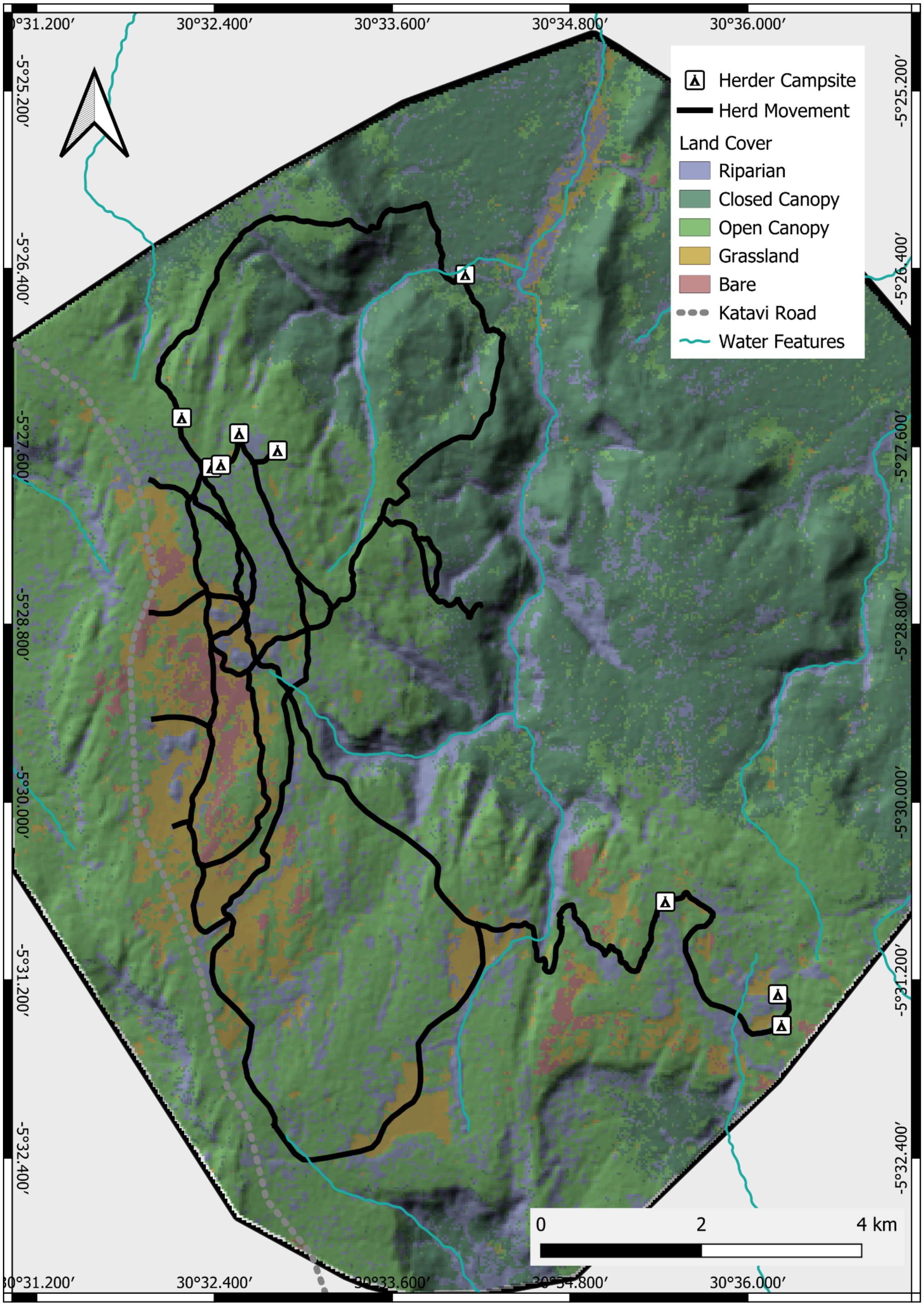
01 — Agropastoral Landscape in the Greater Mahale Ecosystem



A multi-panel locator for field research in the Greater Mahale Ecosystem, western Tanzania. The country inset (bottom right) situates the study within Tanzania; the regional panel (left) shows the Tongwe West Forest Reserve and study villages within the ecosystem that agropastoralists have migrated into; and the detail panel (top right) shows the Issa Valley, where vegetation-survey plots were located across the habitat mosaic and stratified by grazing intensity (control, low, medium, high).

Study villages and plot locations were collected in the field via GPS. Issa Valley habitat classified using a Random Forest model combining optical and radar indices in Google Earth Engine. Layout produced in QGIS.

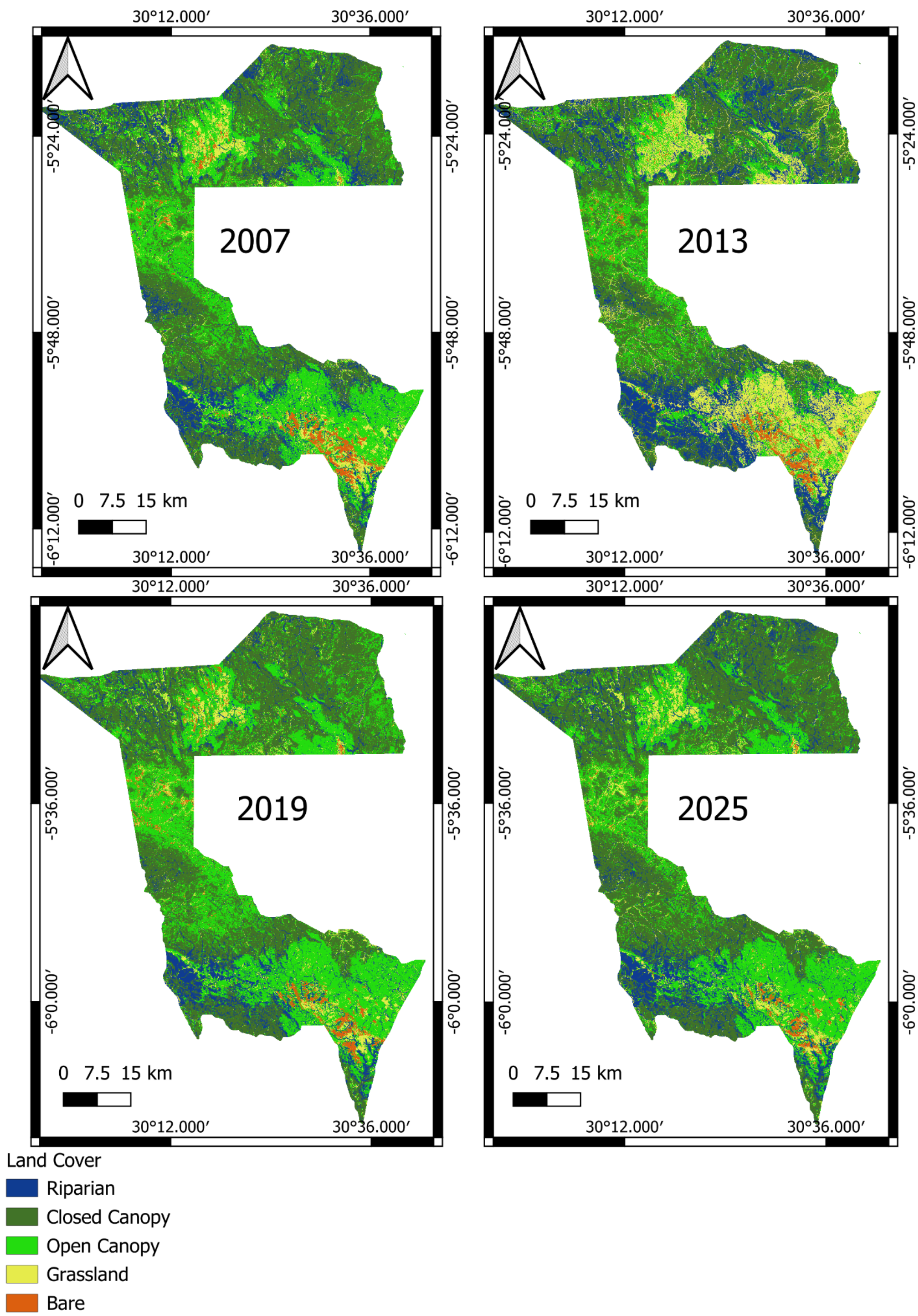
02 — Agropastoral Movement and Use of Landscape



Herder movement and campsites recorded in the Issa Valley, overlaid on habitat and terrain. Campsites cluster in wooded cover while routes track the valley's drainage out to open grazing areas

Herd movement routes and campsites collected in the field via GPS. Overlaid on the Google Earth Engine land-cover classification and a shaded-relief model derived from the Copernicus GLO-30 DEM. Layout produced in QGIS.

03 — Land Cover Change Time Series



A multi-temporal land-cover series for the Tongwe West Forest Reserve, mapping five classes across four epochs (2007–2025), which together reveal a decline in woody cover in 2013 following steady increases in the following years.

Land cover classified per epoch (2007, 2013, 2019, 2025) using a Random Forest model in Google Earth Engine, combining multi-seasonal Landsat optical indices with ALOS PALSAR radar and topographic predictors. Small-multiple layout and shared legend produced in QGIS.